AIRCRAFT RESPONSE ANALYSIS

In order to improve the analytical correlation of aircraft response results, the major portion of a transonic maneuver is divided into a number of segments. In each segment, the buffet pressure data are assumed to be stationary. The aircraft is then subjected to the consecutive application of the buffet loads, and the cumulative dynamic effects are reflected in the time-varying PSD of response. The approach was applied to the following set of flight test data:

Flight Number	871, Run 2
Mach No.	0.925
Altitude	10,668 m (35,000 ft)
Flap Settings	(4°/12°)
Low-Frequency Digital Tape Frame Rate	1000 per sec
Time Increment	0.002 sec
Frequency Increment	0.488 Hz
Spectral Frequency Range	1.4 - 20.0 Hz

Some of the real-time data of this run were given previously in Figures 7-11. Altogether, dynamic pressure data covering the time span 073.00-082.10 second were used. The data were divided into 4 equal time segments. The computation was carried out using the rigid body plunging mode and the first three symmetrical flexible modes. The pitch mode was not included because of lack of tail surface dynamic loads data.

The segment stationary analysis starts with the PSD matrix of the modal forces f_r, f_s where r, s are time segment indices and f_r, f_s are column matrices themselves.

$$\left[S_{f_r f_s}(\omega)\right] = [X]^T \lceil A \rfloor \left[S_{p_r p_s}(\omega)\right] \lceil A \rfloor [X] \tag{7}$$

where

$$\left[S_{p_r p_s}(\omega)\right]_{ij} = \frac{T}{2\pi} F_{p_{ri}}(\omega, T) F^*_{p_{sj}}(\omega, T) \begin{cases} 1 & \text{perfect spatial correlation} \\ \delta_{ij} & \text{zero spatial correlation} \end{cases} \tag{8}$$

The aircraft modal transfer function matrix is $[H(\omega)]$, which is determined based on aircraft mass, and mass distribution, damping, and stiffness data, as well as its aerodynamic characteristics. According to the nonstationary analysis or segment stationary analysis, the PSD matrix of the modal coordinates α due to buffet pressure excitation defined by Eq. (7) is:

$$\begin{aligned}\left[S_\alpha(\omega_1, \omega_2)\right] &= \left[H(\omega_1)\right] \left[S_Q(\omega_1, \omega_2)\right] \left[H^*(\omega_2)\right]^T \\ &= \sum_{r=1}^{n} \sum_{s=1}^{n} \int_{-\infty}^{\infty} A_r(\omega - \omega_1) \left[H(\omega_1)\right] [X]^T \lceil A \rfloor \left[S_{p_r p_s}(\omega)\right] \lceil A \rfloor \\ &\quad [X] \left[H^*(\omega)_2\right]^T A^*_s(\omega - \omega_2)\, d\omega\end{aligned} \tag{9}$$

where $A_r(\omega-\omega_1)$, $A_s(\omega-\omega_2)$ are the Fourier transforms of the deterministic functions defining the buffet pressure inputs in time segments r, s. Introducing a row matrix $\lfloor Y \rfloor$, which is the modal shape matrix, the PSD of deformation w at a specific location of the aircraft and the rms value of $w^2(t)$ may be computed as follows: